



《数值分析》5

—解线性方程组的直接法

主要内容:

方程组化简—消元过程

高斯消元法及算法实现

矩阵初等变换与高斯变换

Frobenius矩阵与LU分解

➤ 线性方程组的矩阵形式

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

.....

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

$$\sum_{j=1}^n a_{ij}x_j = b_i$$

($i=1,2,\cdots,n$)



$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$AX = b$$

$$X? \rightarrow b$$

线性方程组求解:

1. 直接方法;
2. 基本迭代法;
3. 子空间方法

例

$$\begin{cases} x_1 + 2x_2 + x_3 + 4x_4 = 13 \\ 2x_1 + 0x_2 + 4x_3 + 3x_4 = 28 \\ 4x_1 + 2x_2 + 2x_3 + x_4 = 20 \\ -3x_1 + x_2 + 3x_3 + 2x_4 = 6 \end{cases} \quad \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 0 & 4 & 3 \\ 4 & 2 & 2 & 1 \\ -3 & 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 13 \\ 28 \\ 20 \\ 6 \end{bmatrix}$$

$$\begin{cases} x_1 + 2x_2 + x_3 + 4x_4 = 13 \\ -4x_2 + 2x_3 - 5x_4 = 2 \\ -5x_3 - 7.5x_4 = -35 \\ -9x_4 = -18 \end{cases} \quad \begin{bmatrix} 1 & 2 & 1 & 4 \\ & -4 & 2 & -5 \\ & & -5 & -7.5 \\ & & & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 13 \\ 2 \\ -35 \\ -18 \end{bmatrix}$$

$$x_4 = 2, x_3 = 4, x_2 = -1, x_1 = 3$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 4 \\ 2 \end{bmatrix}$$

➤ 解线性方程组的克莱姆方法

1. 输入矩阵 A 和右端向量 b ;
2. 计算 A 的行列式 D , 如果 $D=0$, 则输出错信息结束, 否则进行 3 ;
3. 对 $k=1, 2, \dots, n$ 用 b 替换 A 的第 k 列数据, 并计算替换后矩阵的行列式值 D_k ;
4. 计算并输出 $x_1 = D_1 / D, \dots, x_n = D_n / D$, 结束。

高斯消元法

第一步: 将方程组化简为三角形方程组;

第二步: 解三角形方程组, 获方程组的解。

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \dots\dots\dots \\ a_{nn}x_n = b_n \end{array} \right. \quad (a_{11} \dots a_{nn} \neq 0)$$
$$x_k = [b_k - (a_{k,k+1}x_{k+1} + \dots + a_{k,n}x_n)] / a_{k,k}$$

$$(k = n-1, \dots, 1)$$

除法: n 次; 乘法: $n(n-1)/2$ 次,
乘、除法运算共 $n(n+1)/2$ 次, 简记为 $O(n^2)$

➤消元过程(化一般方程组为上三角方程组)

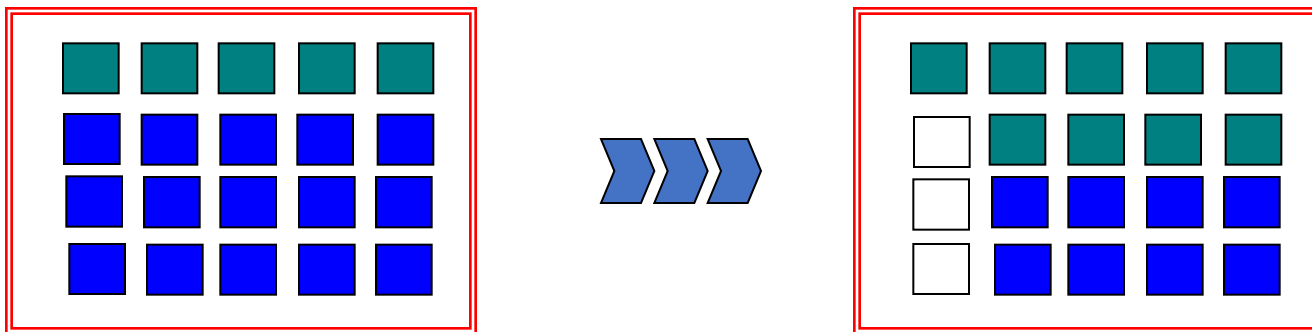
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 = b_3 \\ a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 = b_4 \end{cases}$$

$$\rightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 = b_1 \\ a_{22}^{(1)}x_2 + a_{23}^{(1)}x_3 + a_{24}^{(1)}x_4 = b_2^{(1)} \\ a_{33}^{(2)}x_3 + a_{34}^{(2)}x_4 = b_3^{(2)} \\ a_{44}^{(3)}x_4 = b_4^{(3)} \end{cases}$$

增广矩阵

$$\overline{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & b_1 \\ a_{21} & a_{22} & a_{23} & a_{24} & b_2 \\ a_{31} & a_{32} & a_{33} & a_{34} & b_3 \\ a_{41} & a_{42} & a_{43} & a_{44} & b_4 \end{bmatrix}$$

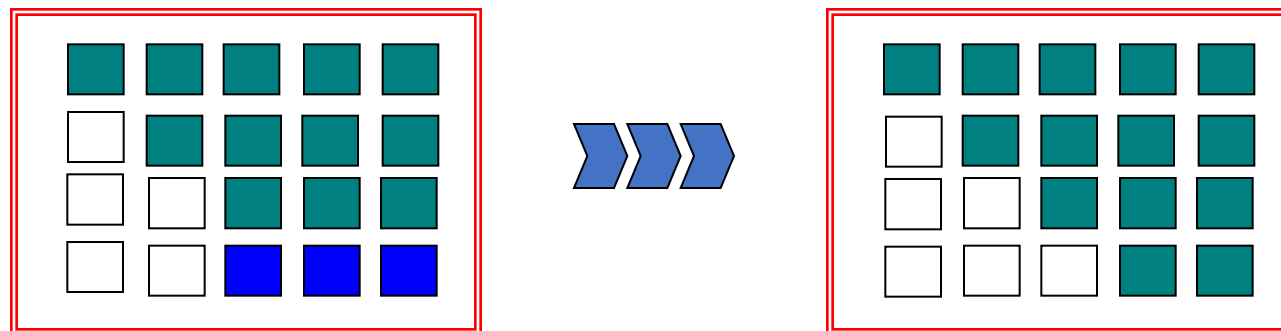
计算: $[m_{21} \ m_{31} \ m_{41}]^T = [a_{21} \ a_{31} \ a_{41}]^T / a_{11}$



实现第一轮消元

$$\overline{A}^{(1)} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & b_1 \\ & a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} & b_2^{(1)} \\ & a_{32}^{(1)} & a_{33}^{(1)} & a_{34}^{(1)} & b_3^{(1)} \\ & a_{42}^{(1)} & a_{43}^{(1)} & a_{44}^{(1)} & b_4^{(1)} \end{bmatrix}$$

$$\text{计算: } [m_{32} \ m_{42}]^T = [a_{32}^{(1)} \ a_{42}^{(1)}] / a_{22}^{(1)}$$



上三角方程组

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ & a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ & & a_{33}^{(2)} & a_{34}^{(2)} \\ & & & a_{44}^{(3)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2^{(1)} \\ b_3^{(2)} \\ b_4^{(3)} \end{bmatrix}$$

n 阶方程组消元过程乘法次数:

$$(n-1)n + (n-2)(n-1) + \dots + 1 \times 2 = (n^3 - n)/3$$

除法次数: $(n-1) + (n-2) + \dots + 1 = n(n-1)/2$

回代过程: $n(n+1)/2$ 总: $n^2 + (n^3 - n)/3$, 简记 $O(n^3)$

n	2	3	4	5	6
高斯	6	17	36	65	106
克莱姆	8	51	364	2885	25206

高斯消元法算法:

1. **For** $k=1, \cdots, n-1$ **Do**
2. **For** $i=k+1, \cdots, n$ **Do**
3. $a_{ik} \leftarrow a_{ik} / a_{kk}$
4. **For** $j=k+1, \cdots, n$ **Do**
5. $a_{ij} \leftarrow a_{ij} - a_{ik} \times a_{kj}$
6. **EndDo**
7. $b_i \leftarrow b_i - a_{ik} \times b_k$
8. **EndDo**
9. **EndDo**

➤ 初等矩阵与Gauss变换

$$\begin{aligned}
 I = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}_{n \times n} & \xrightarrow[\text{加到第二行}]{\text{第一行乘 } -m_{21}} E_2 = \begin{bmatrix} 1 & & & \\ -m_{21} & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}_{n \times n} \\
 E_3 = \begin{bmatrix} 1 & & & \\ & 1 & & \\ -m_{31} & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix}_{n \times n} & \dots \dots \dots E_n = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ -m_{n1} & & & & 1 \end{bmatrix}_{n \times n} \\
 F_1 = E_n \cdots E_3 E_2 I & \Rightarrow F_1 = \begin{bmatrix} 1 & & & \\ -m_{21} & 1 & & \\ \vdots & & \ddots & \\ -m_{n1} & & & 1 \end{bmatrix}_{n \times n}
 \end{aligned}$$

$$F_1 = E_n \cdots E_3 E_2$$

第一轮消元: $A^{(1)} = F_1 A$

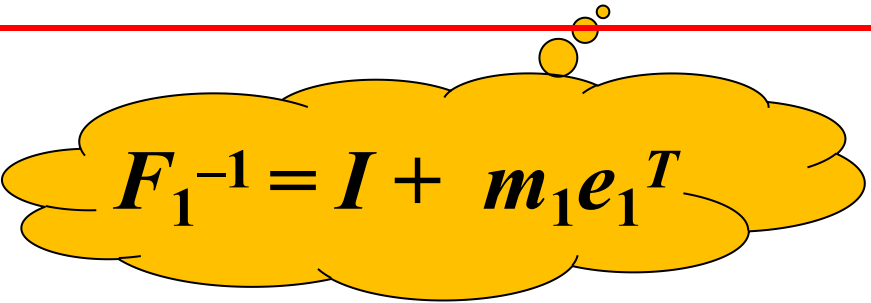
$$A^{(1)} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -m_{21} & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ -m_{n1} & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$
$$= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{22}^{(1)} & \cdots & a_{2n}^{(1)} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} \end{bmatrix} \quad F_1^{-1} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ m_{21} & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ m_{n1} & 0 & \cdots & 1 \end{bmatrix}$$



关键：如何去找 F_1

$$\begin{bmatrix} 0 & 0 & \cdots & 0 \\ m_{21} & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ m_{n1} & 0 & \cdots & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ m_{21} \\ \vdots \\ m_{n1} \end{bmatrix} [1 \ 0 \ \cdots \ 0] = m_1 e_1^T$$

$$F_1 = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -m_{21} & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ -m_{n1} & 0 & \cdots & 1 \end{bmatrix} = I - m_1 e_1^T$$


$$F_1^{-1} = I + m_1 e_1^T$$

$$F_k = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & -m_{k+1,k} & & 1 & \\ & \vdots & & & \ddots \\ & -m_{nk} & & & & 1 \end{bmatrix} = I - m_k e_k^T \quad (k = 1, 2, \dots, n-1)$$

$$m_k = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ m_{k+1,k} \\ \vdots \\ m_{nk} \end{bmatrix} \quad e_k^T = [0 \cdots 0 \ 1 \ 0 \cdots 0]$$

$$F_k^{-1} = I + m_k e_k^T$$

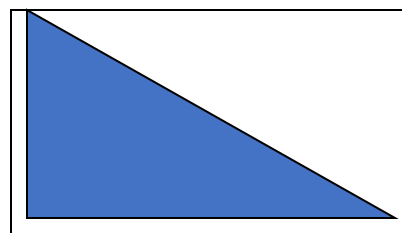
➤ Gauss消元结果

$$A^{(n-1)} = F_{n-1} F_{n-2} \cdots F_1 A$$

$$F_k = I - m_k e_k^T \quad (k = 1, 2, \cdots, n-1)$$

称 F_k 为Frobenius矩阵, $F_k^{-1} = I + m_k e_k^T$

$$F_1^{-1} F_2^{-1} \cdots F_{n-1}^{-1} =$$



$$\begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & m_{n,n-1} & 1 \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} 1 & & & \\ m_{21} & \ddots & & \\ \vdots & \ddots & 1 & \\ m_{n1} & \cdots & m_{n,n-1} & 1 \end{bmatrix}$$

$$F_{n-1}F_{n-2}\cdots\cdots F_1 A = A^{(n-1)} \quad \rightarrow$$

$$A = \underbrace{(F_1^{-1}F_2^{-1} \cdots F_{n-1}^{-1})}_L \underbrace{A^{(n-1)}}_U$$

$$L = \begin{bmatrix} 1 & & & \\ m_{21} & \ddots & & \\ \vdots & \ddots & & 1 \\ m_{n1} & \cdots & m_{n,n-1} & 1 \end{bmatrix} \quad U = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} \\ & & \ddots & \cdots \\ & & & a_{nn}^{(n-1)} \end{bmatrix}$$

矩阵的三角分解: $A = LU$

举例 用高斯消元法对如下系数矩阵A进行三角分解

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 5 & 2 \\ 4 & 3 & 30 \end{bmatrix} \xrightarrow[\text{(-2)r}_1+\text{r}_3]{\text{(-3/2)r}_1+\text{r}_2} F_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{4}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow A^{(1)} = F_1 A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & \frac{1}{2} & -4 \\ 0 & -3 & 22 \end{bmatrix} \xrightarrow{6\text{r}_2+\text{r}_3} F_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{3}{0.5} & 1 \end{bmatrix}$$

$$\rightarrow A^{(2)} = F_2 A^{(1)} = \begin{bmatrix} 2 & 3 & 4 \\ 0 & \frac{1}{2} & -4 \\ 0 & 0 & -2 \end{bmatrix} \rightarrow F_2 F_1 A = A^{(2)}$$

$$\rightarrow A = (F_2 F_1)^{-1} A^{(2)} = \boxed{F_1^{-1} F_2^{-1}} \boxed{A^{(2)}} \begin{matrix} \text{L} \\ \text{U} \end{matrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 0 & 4 & 3 \\ 4 & 2 & 2 & 1 \\ -3 & 1 & 3 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & -4 & 2 & -5 \\ 4 & -6 & -2 & -15 \\ -3 & 7 & 6 & 14 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & -4 & 2 & -5 \\ 4 & 3/2 & -5 & -7.5 \\ -3 & -7/4 & 19/2 & 21/4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & -4 & 2 & -5 \\ 4 & 3/2 & -5 & -7.5 \\ -3 & -7/4 & -19/10 & -9 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & & & \\ 2 & 1 & & \\ 4 & 3/2 & 1 & \\ -3 & -7/4 & -19/10 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 2 & 1 & 4 \\ & -4 & 2 & -5 \\ & & -5 & -7.5 \\ & & & -9 \end{bmatrix}$$

思考与练习

1. 分析Frobenius矩阵 $F_k = I - m_k e_k^T$ 的结构,证明 $F_k^{-1} = I + m_k e_k^T$ 为其逆矩阵。
2. 证明“下三角矩阵的逆矩阵也是下三角矩阵”是否正确
3. 证明“两个下三角矩阵的乘积矩阵也是下三角矩阵”
2. 证明 $(F_1^{-1} F_2^{-1} \cdots F_{n-1}^{-1})$ 是下三角矩阵

学到了什么？



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